



Daffodil
International
University

Lab Report

Course Code : CSE-322

Course Title : Computer Networks Lab

Experiment No : 01

Experiment Name : DHCP Configuration and Implementation in a Simulated Network Setup.

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Submission Date : 16/03/2025

Experiment No. 01

Experiment Name:

DHCP Configuration and Implementation in a Simulated Network Setup.

Objectives:

To understand the purpose and working mechanism of DHCP in networking, including its role in IP address allocation and network configuration. To configure and verify DHCP services using Cisco Packet Tracer, ensuring that devices receive proper IP addresses dynamically. To analyze the advantages and disadvantages of DHCP in network management and understand how it impacts network efficiency and scalability.

Theory:

The Dynamic Host Configuration Protocol (DHCP) is a network management protocol that automates the assignment of IP addresses, subnet masks, gateways, and DNS configurations to devices within a network. It eliminates the need for manual IP configuration, thereby reducing administrative overhead and minimizing errors. DHCP follows a four-step process known as DORA (Discover, Offer, Request, Acknowledge) to dynamically assign IP addresses to client devices. This protocol is widely used in home networks, corporate environments, and large-scale enterprise networks, ensuring seamless device connectivity.

Importance of DHCP in Networking:

Automation of IP Assignment: DHCP automatically assigns IP addresses to devices, reducing human errors and ensuring efficient network management. Manual IP configuration in large networks is time-consuming and prone to errors, making DHCP a crucial solution.

Scalability for Growing Networks: DHCP can handle networks of varying sizes, from small office setups to large enterprise environments with thousands of devices. It allows new devices to join the network seamlessly without the need for manual intervention.

Efficient Utilization of IP Addresses: DHCP dynamically assigns and reclaims IP addresses when they are no longer in use, preventing IP address exhaustion and optimizing network resources.

Ensures Network Consistency: By automatically assigning the correct subnet mask, gateway, and DNS settings, DHCP ensures that all devices in the network have a uniform configuration, reducing configuration mismatches.

How DHCP Works: The DORA Process:

Discover: When a device connects to the network, it sends a DHCP Discover broadcast message to locate an available DHCP server. Since the client does not yet have an IP address, this message is broadcast across the network.

Offer: A DHCP server responds with a DHCP Offer message, suggesting an available IP address along with network configuration details such as subnet mask, default gateway, and DNS servers.

Request: The client selects one of the offered addresses and sends a DHCP Request message to confirm its selection and request the IP address from the DHCP server.

Acknowledge: The DHCP server sends a DHCP Acknowledgment (ACK) message, confirming the lease and officially assigning the IP address to the client. The client can now use the IP address for network communication.

Components of DHCP Configuration:

DHCP Server: A centralized system responsible for managing and assigning IP addresses to clients based on predefined rules. It maintains an IP address pool and handles address allocation requests from clients.

DHCP Scope: A predefined range of IP addresses that the DHCP server is allowed to assign to client devices. The scope defines the available pool of IP addresses for dynamic allocation.

Lease Time: The duration for which an IP address is assigned to a client before it must be renewed. Shorter lease times are ideal for dynamic networks, while longer lease times are suitable for stable networks.

Subnet Mask: Defines the IP range of a network, allowing devices to determine which IP addresses belong to the same local network and which require routing to external networks.

Default Gateway: The IP address of the network router that connects internal devices to external networks, enabling internet access and communication beyond the local network.

DNS Servers: Translate human-readable domain names (e.g., www.example.com) into numerical IP addresses, allowing devices to access websites and online resources efficiently.

DHCP Configuration in Cisco Packet Tracer:

Define the IP Pool: Configure the DHCP server with an IP pool, specifying the range of addresses available for allocation, along with subnet mask, default gateway, and DNS server settings.

Verify Client Configuration: Connect multiple client devices to the network and enable DHCP to check if they receive unique IP addresses automatically from the DHCP server.

Troubleshoot DHCP Issues: Diagnose common DHCP-related issues such as IP conflicts, incorrect configurations, or connectivity failures. Verify server settings and client configurations to ensure smooth operation.

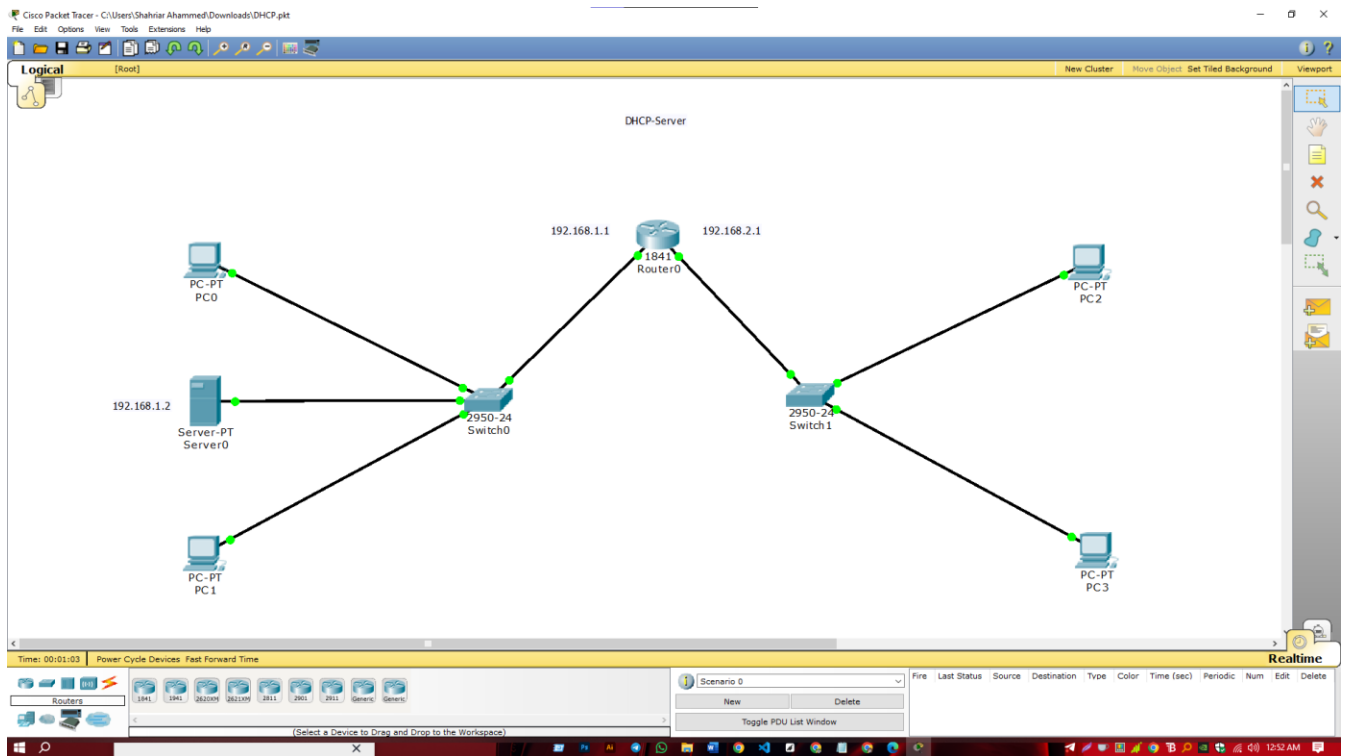


Fig 01: Setup of DHCP Server

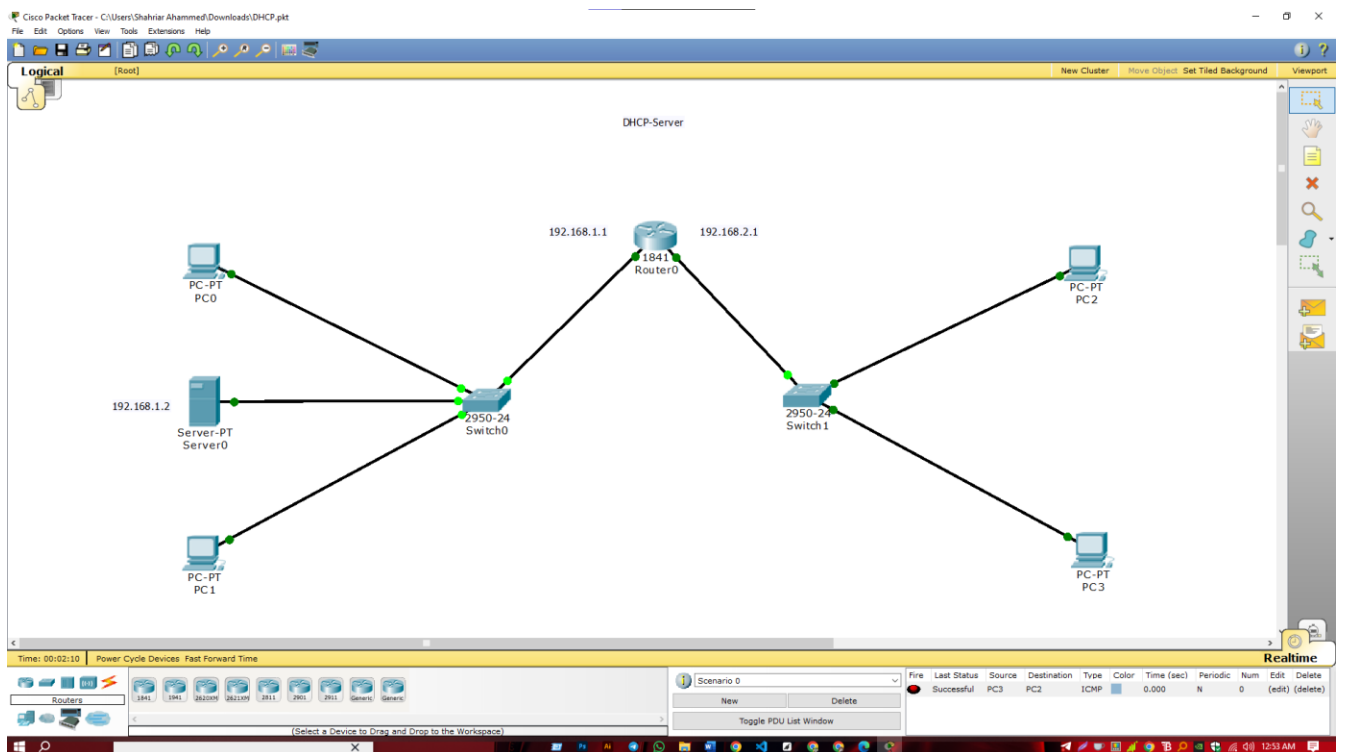


Fig 02: Sending Message Successful

Advantages of DHCP:

Automated IP Address Management: Reduces administrative workload by dynamically assigning IP addresses without manual intervention.

Efficient Use of Network Resources: Ensures optimal allocation and utilization of available IP addresses, preventing address wastage.

Scalability: Adapts to the needs of both small and large networks, handling thousands of IP assignments effortlessly.

Simplified Network Setup: New devices can join the network without requiring manual IP configuration, making the onboarding process easier.

Consistent and Reliable Configuration: Provides uniform network settings across all connected devices, ensuring a stable and predictable network environment.

Disadvantages of DHCP:

Dependency on DHCP Server Availability: If the DHCP server fails, devices will be unable to obtain IP addresses, causing network disruptions. Redundant DHCP servers can mitigate this risk.

Security Vulnerabilities: Rogue DHCP servers can assign incorrect IP configurations, leading to potential security risks such as man-in-the-middle attacks. Proper security measures should be implemented.

IP Address Exhaustion: In networks with a limited DHCP scope, there may be more client devices than available IPs, resulting in connectivity issues. Proper planning is essential to avoid this problem.

Reduced Control Over Static IP Assignments: Since DHCP dynamically assigns IP addresses, manually assigning specific addresses to critical devices (such as servers) becomes challenging. Static IP reservations are required for such cases.

Potential IP Conflicts: If multiple DHCP servers exist on the same network or if static IPs are assigned within the DHCP scope, conflicts may arise, disrupting network stability.

Practical Implementation in Cisco Packet Tracer:

Network Topology Setup: Create a simulated network in Cisco Packet Tracer, including a DHCP server (or router), a switch, and multiple client devices such as PCs or laptops.

DHCP Server Configuration: Define the DHCP pool, including the range of IP addresses, subnet mask, default gateway, and DNS settings. Ensure that all network parameters are correctly set.

Client Verification: Enable DHCP on all client devices and check whether they receive unique IP addresses dynamically from the server.

Troubleshooting and Optimization: Diagnose any potential issues, such as incorrect DHCP scope settings, connectivity problems, or lease expiration problems, and take corrective actions.



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Lab Report

Course Code : CSE-322

Course Title : Computer Networks Lab

Experiment No : 02

Experiment Name : Configuration and Implementation of RIP Version 1 and Version 2 in a Simulated Network Environment.

Submitted To : Ms. Chayti Saha (Lecturer)

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Submission Date : 16/03/2025

Experiment No. 02

Experiment Name:

Configuration and Implementation of RIP Version 1 and Version 2 in a Simulated Network Environment.

Objectives:

To understand the purpose and operational mechanisms of RIP in networking, including how it facilitates dynamic routing.

To configure and verify RIP Version 1 and Version 2 on network devices using Cisco Packet Tracer.

To explore the key differences between RIP Version 1 and Version 2, focusing on classful and classless routing.

Theory:

The Routing Information Protocol (RIP) is a distance-vector routing protocol that determines the best path to a destination using the hop count metric. Each router running RIP periodically broadcasts its entire routing table to its neighbors, allowing them to update their tables accordingly. The protocol operates on the Bellman-Ford Algorithm, which calculates the shortest path based on hop count.

RIP Version 1 (RIPv1): A classful routing protocol that does not support subnet mask information in updates, limiting its efficiency in modern networks.

RIP Version 2 (RIPv2): An improved version that supports classless routing, including Variable Length Subnet Masking (VLSM) and authentication for enhanced security.

RIP uses a maximum hop count of 15, meaning any destination beyond 15 hops is considered unreachable, preventing routing loops but restricting its scalability.

Importance of RIP in Networking:

Simple Configuration: RIP is easy to set up and manage, making it suitable for small to medium-sized networks with basic routing needs.

Device Compatibility: It is supported by a wide range of networking devices, ensuring interoperability in mixed vendor environments.

Automatic Route Updates: RIP automatically updates its routing table at fixed intervals (every 30 seconds), reducing the need for manual intervention.

Loop Prevention Mechanisms: Features like split horizon and poison reverse help prevent routing loops, ensuring stable network performance.

How RIP Works:

Route Advertisement: Each RIP-enabled router shares its routing table with its directly connected neighbors at periodic intervals.

Hop Count as Metric: The number of routers a packet passes through is counted as hops. The fewer the hops, the better the route.

Routing Updates: Routers update their routing tables based on received information and propagate the changes throughout the network.

Route Aging & Timeouts: If a route is not updated within a specific period, it is considered unreachable and removed from the routing table.

Components of RIP Configuration:

RIP Version: Determines whether RIPv1 (classful) or RIPv2 (classless) is being used.

Network Statements: Define which networks should be advertised in RIP updates.

Timers: Control update frequency and route expiration, including update, invalid, hold-down, and flush timers.

Authentication (RIPv2 Only): Ensures that only trusted routers exchange routing information, enhancing security.

RIP Configuration in Cisco Packet Tracer:

Define the Network Topology: Create a simulated network with multiple routers and subnets.

Configure RIP on Routers: Enable RIP, select the version, and specify the networks to be advertised.

Verify Configuration: Check routing tables on routers to ensure proper route learning and propagation.

Troubleshoot Issues: Identify and resolve problems such as missing network statements, incorrect RIP versions, and routing loops.

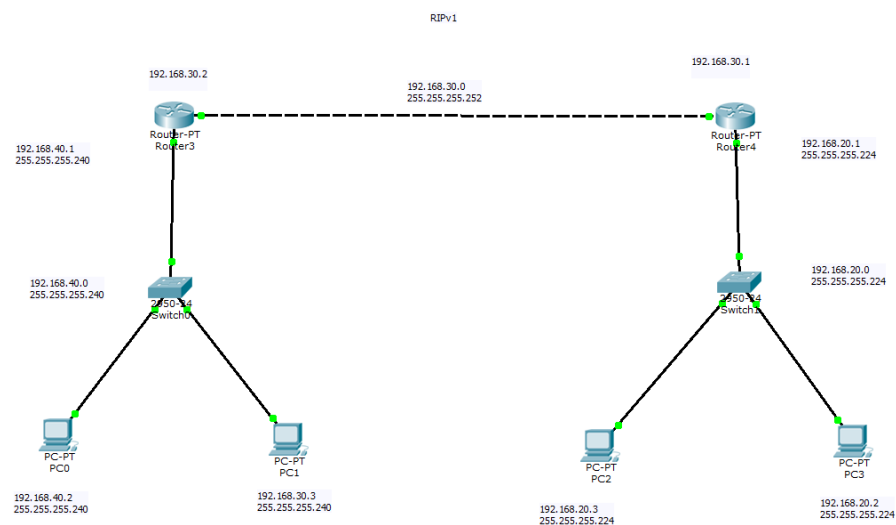


Fig 1: Set up of RIP Version 1

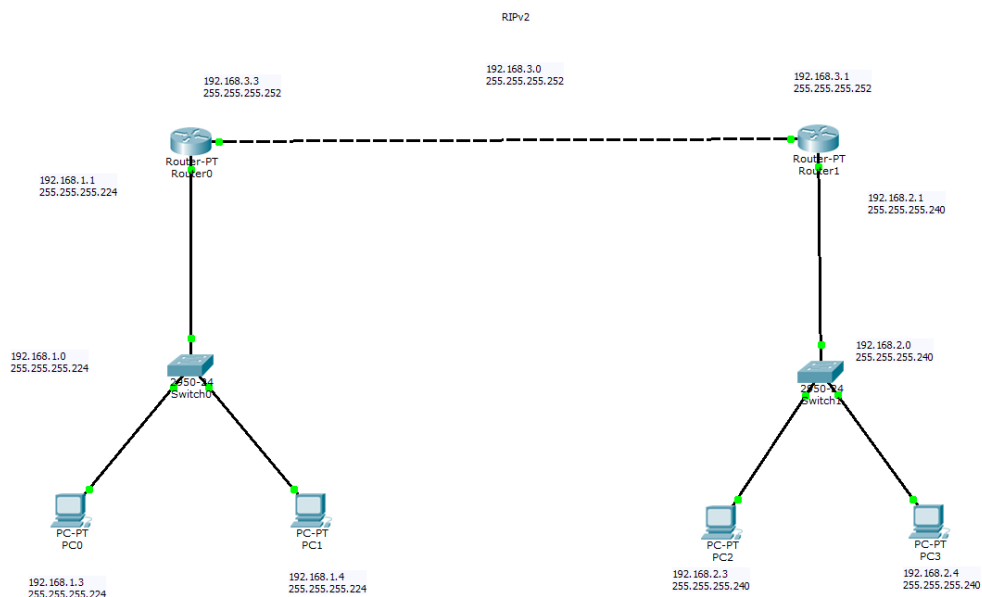


Fig 2: Set up of RIP Version 2

Advantages of RIP:

Ease of Deployment: Simple configuration and maintenance make it ideal for beginner network administrators.

Automated Updates: Routers exchange routing information periodically without manual input.

Widespread Support: Compatible with various networking devices from different vendors.

Loop Prevention: Implements strategies like split horizon and poison reverse to avoid routing loops.

Disadvantages of RIP:

Limited Scalability: Maximum hop count of 15 restricts its use in large networks.

Slow Convergence: Routing updates take time to propagate, causing potential delays in network adjustments.

High Bandwidth Usage: Frequent routing table broadcasts consume network bandwidth.

Lack of VLSM Support (RIPv1): RIPv1 does not support subnet masks, leading to inefficient IP address utilization.

Practical Implementation in Cisco Packet Tracer:

Setup: Design a network topology with multiple routers and subnets.

Enable RIP: Configure RIP on routers and define network statements.

Verify Configuration: Use commands like `show ip route` and `show ip protocols` to check route updates.

Troubleshooting: Fix common RIP-related issues such as missing routes or incorrect version settings.